CS 2002: Design and Analysis of Algorithms (Test -1)

Name and Roll No:

1. Consider the QuickSort algorithm discussed in class. Suppose that, instead of choosing the exact median as the pivot, we choose an approximate median in $O(n)$ time such that, after partitioning an array of n elements, the size of each of the two resulting subproblems is at least $n/4$ and at most $3n/4$.

   (a) Determine an appropriate recurrence relation for the running time of QuickSort under this pivot-selection strategy. Clearly justify your recurrence based on the given guarantee on the subproblem sizes. (1 Mark)

   (b) Solve the recurrence obtained in part (a) and determine the asymptotic running time of the resulting QuickSort algorithm. (1 Mark)

2. Consider the problem of finding the closest pair of points among a given set of n points in the plane given in an array Arr. In the divide-and-conquer algorithm discussed in the class, the problem is divided into two subproblems using a median position in this ordering. Suppose that the point at the median position has x-coordinate x_d, and that more than $n/2$ points have the same x-coordinate x_d. Note that we not sort Arr to find d. Answer the following questions.

   (a) Suppose that, in order to obtain two subproblems of roughly equal size, we assign the points (in Arr) having x-coordinate $\leq x_d$ to the left subproblem and the remaining points to the right subproblem. Discuss carefully what happens at the divide, conquer, and combine steps, and identify any difficulties that arise. (1.5 Mark)

   (b) How would you sort out the issues of (a) if any, by investing time linear to the size of the problem? (1.5 Mark)

3. Consider a binary tree with n nodes. Answer the following questions:

   (a) Design a recursive algorithm to perform an inorder traversal of the given binary tree. (1 Mark)

   (b) Based on your algorithm, formulate a recurrence relation for its running time. Clearly explain the contribution of each recursive call and the work performed at each node. (1 Mark)

   (c) Solve the recurrence relation and determine the asymptotic running time of the algorithm using Big-O notation. Explain the steps clearly. (1 Mark)

4. (a) Student A was solving a Dynamic Programming problem involving two strings $X = x_1x_2 \dots x_m$ and $Y = y_1y_2 \dots y_n$. Unfortunately, Student B lost the question paper and was able to recover only the following recurrence:

$$DP[i][j] = \begin{cases} 0, & \text{if } i = 0 \text{ or } j = 0, \\ DP[i-1][j-1] + 1, & \text{if } x_i = y_j, \\ \max\{DP[i-1][j], DP[i][j-1]\}, & \text{if } x_i \neq y_j. \end{cases}$$

Help Student B reconstruct the missing question. In particular:

i. What problem is Student A most likely solving? (1.5 Mark)

ii. What does $DP[i][j]$ represent? (1.5 Mark)